

ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems.

Experiment: 10- 5G Network Slicing Environment Using MATLAB.

Objective – 1: Analyze the Number of Network Slices:

Program:

```
clc;

clear;

close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};

count = [1 1 1 1 1]; % One instance of each network slice

figure

bar(count)

grid on

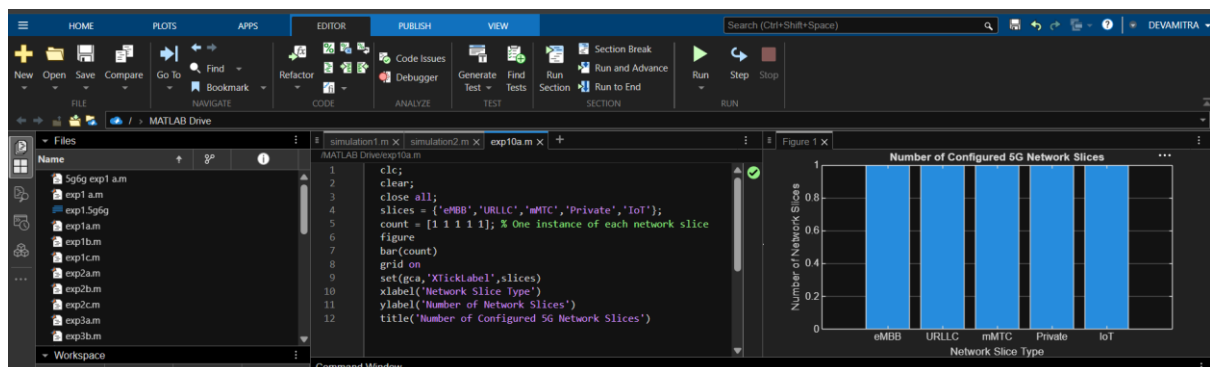
set(gca,'XTickLabel',slices)

xlabel('Network Slice Type')

ylabel('Number of Network Slices')

title('Number of Configured 5G Network Slices')
```

Output:



Objective – 2 :Compare Slice Utilization (%):

Program:

```
clc;

clear;

close all;

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});

utilization = [85 70 60 75 50];

figure;

bar(slices, utilization);

grid on;

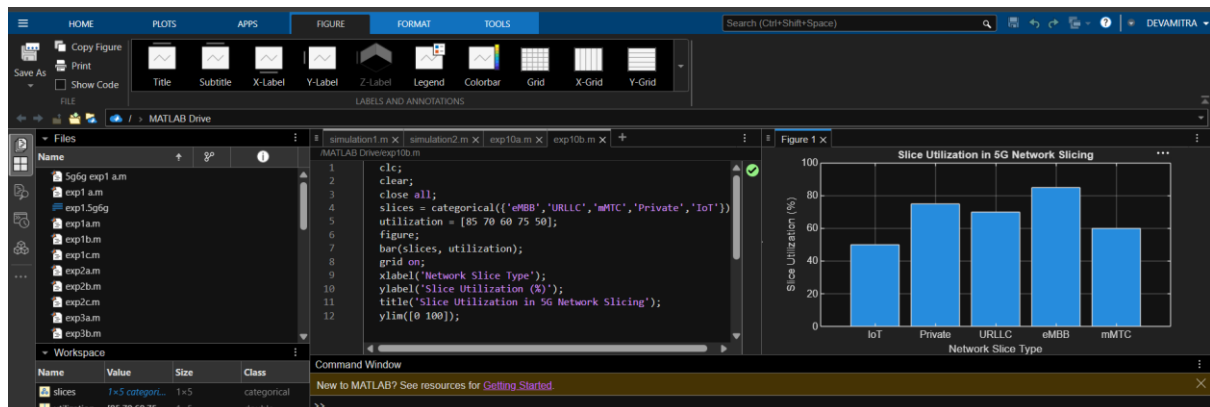
xlabel('Network Slice Type');

ylabel('Slice Utilization (%)');

title('Slice Utilization in 5G Network Slicing');

ylim([0 100]);
```

Output:



Objective – 3 :Analyze End-to-End Latency (ms):

Program:

```
clc;

clear;

close all;

slices = [1 2 3 4 5];

latency = [25 5 40 15 60];

figure;

plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);

grid on;

xticks(slices);

xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});

xlabel('Network Slice Type');

ylabel('End-to-End Latency (ms)');

title('5G Network Slicing: End-to-End Latency');

ylim([0 70]);
```

Output:

